Tracked but Suppressed: How LLMs Fail Current-Value Retrieval

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Abstract

Language models pretrained on trillions of tokens cannot reliably retrieve the current value of a variable that has been overwritten multiple times in their own input — a skill that underpins multi-turn dialogue, retrieval-augmented systems, and tool-augmented agents. Across 20 pretrained models (14 open-weight from 270M to 9B parameters, 6 proprietary frontier), this failure manifests as a general loss of position-indexed retrieval: consistently higher accuracy on the first value than on the current value (mean gap +27pp, range +6 to +53pp at representative cells), and near-zero accuracy on intermediate positions of the update stream. The failure is not architectural. A 7.4-million-parameter LoRA — 0.24% of one model's weights — restores accuracy to 93–100% on harder held-out cells; a training-free format change produces similar recovery. Across 79 published checkpoints from two model families, the failure is present at the earliest pretraining checkpoint we can evaluate and survives every post-training stage. The capability is tracked in pretrained models but suppressed downstream — present in mid-layer representations, absent from the output. Mechanistically, LoRA amplifies an attention-head circuit that already attends partially to the latest value in the pretrained model and installs downstream redundancy that absorbs upstream perturbations.

Stream (K=3, N=3, randomly interleaved):

```
color: red   size: large   shape: round
size: small   color: blue   shape: square
shape: triangle   color: green   size: medium
```

FVQ: What was the *first* value of `color`? → **red**
CVQ: What is the *current* value of `color`? → **green**

Figure 1: Task illustration (K=3, N=3). The stream randomly interleaves updates for all K keys. To answer CVQ correctly, a model must isolate `color` entries from competing keys and identify the most recent one among them.

1 Introduction

Language models are increasingly deployed in settings where the value of a variable changes over time and the model must use the current value, not an earlier one. Multi-turn assistants must respect the most recent user correction; retrieval-augmented systems must use the latest version of a document; tool-augmented agents must read the current contents of a variable after a sequence of writes. Failures of this skill have been documented at the application level — Laban et al. (2025) report substantial degradation in multi-turn dialogues as state updates cascade — but the underlying mechanism, and whether the failure is intrinsic to current models, is not understood.

Figure 1 illustrates the task: K interleaved keys each receiving N updates, with the model asked for either the first (FVQ) or current (CVQ) value of a target key (formal setup in §3). Both queries are trivially solvable from context, so failure indicates a retrieval problem, not a length or memory limit.

We evaluate 20 pretrained models spanning 14 open-weight checkpoints (270M–9B parameters) and 6 proprietary frontier models. At a representative cell per model block, FVQ−CVQ gaps range from +6 to +53 percentage points, with the majority exceeding +20 pp (Table 1, Figure 2). The gap takes two shapes: **primacy** (FVQ exceeds CVQ; majority of cells) and **reversal** (CVQ exceeds FVQ but both below ceiling; smaller open-weight models at high load). When we ask for any intermediate position k of a target key ($1 < k < N$), baseline accuracy drops below 5 % within two positions of the stream start (Figure 3; full sweep across six cells in Appendix F), indicating a general loss of position-indexed retrieval rather than a first-vs-current confusion.

The failure is not an architectural limit: a small 0.24%-parameter LoRA and a training-free format change both restore accuracy to 93–100% on harder held-out cells.

We make the following contributions:

- **Behavioural characterisation:** FVQ–CVQ gap across 20 models, including a previously-undocumented reversal regime (§4).
- **Two recoveries:** task-specific LoRA and training-free format change both close the gap (§5).
- **Training-dynamics analysis** across 79 checkpoints from two model families; failure persists at every documented stage (§6).
- **Mechanism:** LoRA amplifies a partial v_{last}-attending circuit already present in pretraining and installs downstream redundancy; signatures replicate on Gemma-3-4b-it (§7).

2 Related Work

Retrieval failures in LLMs. Application-level work documents that LLMs struggle as in-context state changes: multi-turn dialogues degrade as state updates cascade (Laban et al., 2025); entity-state tracking is unevenly present across models (Kim and Schuster, 2023; Kim et al., 2024; Rezaee et al., 2025); and most closely, Wang and Sun (2025) construct a 46-key stream with up to 400 updates and a single current-value query, reporting a log-linear accuracy decline they conclude is fundamental. The position-dependent failure parallels the static-context "lost in the middle" pattern (Liu et al., 2024), here arising under in-context updates rather than retrieval from a fixed document. Format sensitivity can shift accuracy by up to 76 percentage points (Sclar et al., 2024); our format intervention is a controlled instance. Knowledge-editing methods (Meng et al., 2022, 2023) address an adjacent failure — updating *parametric* facts stored in weights — whereas our setting requires tracking *in-context* updates without any weight change. The task structure parallels working-memory updating paradigms in cognitive psychology (Daneman and Carpenter, 1980), where the operative skill is reading the most recent assignment to a slot rather than recalling all earlier ones. The shared pattern across this line of work is to characterise the failure; we show that this failure is recoverable from the pretrained architecture by either a 0.24%-parameter LoRA or a training-free format change, and trace the recovery to specific attention-head circuits.

Fine-tuning surfaces latent capacity. Recent work has shown that capabilities not visible in pretrained models can be surfaced with minimal supervision. Zhou et al. (2023) demonstrate that 1,000 curated dialogues are sufficient to teach a pretrained model to follow chat-style instructions, attributing this to pretraining having already done the work of knowledge acquisition. Sun and Dredze (2025) make a more general claim across 18 datasets: pretraining improves models in latent ways only visible after fine-tuning. We use standard low-rank adaptation (Hu et al., 2022; Aghajanyan et al., 2021). Our contribution against this background is the concrete instance with mechanism: a specific capability that pretrained LLMs systematically fail at, surfaced by an 18,000-example LoRA across held-out grid cells, datasets, and $\sim$10$\times$ context lengths, with the recovery traced mechanistically to upper-middle-layer attention heads.

Mechanism and pretraining dynamics. Mechanistic interpretability has identified circuits for specific capabilities (Olsson et al., 2022; Wang et al., 2023; Conmy et al., 2023; Li et al., 2025), including how LMs bind entities to attributes and retrieve those bindings (Feng and Steinhardt, 2024; Gur-Arieh et al., 2025); our task isolates failure of the positional retrieval path under destructive in-context updates. Closest to our setting, Mishra (2025) report that summarization fine-tuning shifts the function of 62% of attention heads; we apply the same approach to a mechanistic capability rather than a style task. On the pretraining-capability side, Wei et al. (2022) and Schaeffer et al. (2023) debate whether abilities emerge at scale; our work is a *non-emergence* question — this skill does not emerge at any scale we test. Allen-Zhu and Li (2023) parallel our claim for memorised knowledge: pretrained models store the information but do not develop a sufficient retrieval pathway by default. We add the converse mechanistic finding for an in-context capability: when the retrieval pathway is surfaced by a small fine-tune, it operates by amplifying a partial circuit already present in pretraining.

3 Task and Setup

Task. Let $\mathcal{K} = \{k_1, \ldots, k_K\}$ be a set of keys and let $V_i = (v_{i,1}, \ldots, v_{i,N})$ be an ordered update sequence for key k_i. The input stream σ is a random interleaving of all $K \times N$ pairs $\{(k_i, v_{i,j})\}$, subject to the constraint that no two consecutive entries share the same key. The model receives σ and a target key k_t and is asked one of three queries: **FVQ** (first-value, expected answer $v_{t,1}$); **CVQ** (current-value, expected answer $v_{t,N}$); or **IVQ** (intermediate-value, expected answer $v_{t,k}$ for some $1 < k < N$). Figure 1 shows a small instance. The intermediate-value query is used only in the dense-retrieval analysis (§5; Appendix F).

Datasets. **Arbitrary-Single (ARB)** uses a pool of 2,300 single-token English words across 46 semantic categories; single-token values keep cross-entropy and probing computations clean and make it the dataset for the mechanistic analysis (§7). **Semantic-Multi (SEM)** uses real category members as multi-token values across the same 46 categories (2,403 entries total) and is our primary behavioural dataset — semantic coherence between key and value limits template-matching shortcuts. Full construction details are in Appendix A.

Grid and trials. We evaluate models on a unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$, with up to 200 trials per (K, N) cell per query type. Cells where the prompt exceeds a model's context limit or where the dataset pool cannot supply N distinct values per key are excluded (specifics in Appendix A).

Metric. For each (K, N) cell we report FVQ and CVQ accuracy as string-match proportions (matching rule in Appendix B) with Wilson 95% confidence intervals (Wilson, 1927). The primary metric is the **gap** Δ = FVQ accuracy − CVQ accuracy; we call a gap **substantial** when $|\Delta| > 0.05$. All inferences use greedy decoding (temperature 0) to eliminate sampling variance.

Models. We evaluate **11 open-weight models** from three major families — Qwen2.5 (0.5B–3B; instruct and base), Qwen3.5 (0.8B–9B; instruct), Gemma-3 (270M–4B; instruct) — and **6 proprietary frontier models** from OpenAI (GPT-4.1, GPT-4.1-mini), Anthropic (Claude-4.5-Haiku, Claude-4.5-Sonnet), and Google (Gemini-2.5-Flash, Gemini-2.5-Pro). Three additional smaller models — TinyLlama-1.1B, StableLM-2-1.6B, Pythia-410M — appear only in the appendix. The full model list is in Appendix C; per-cell behavioural results are in Appendix D.

Model	FVQ	CVQ	Gap	% Ctx
Proprietary models, K=20, N=50				
GPT-4.1	1.00	0.71	+0.29	0.7%
GPT-4.1-mini	1.00	0.74	+0.26	0.7%
Claude-4.5-Haiku	1.00	0.53	+0.47	3.3%
Claude-4.5-Sonnet	1.00	0.74	+0.26	0.7%
Gemini-2.5-Flash	1.00	0.54	+0.46	0.6%
Gemini-2.5-Pro	0.90	0.77	+0.13	0.7%
Open-weight models, K=5, N=30				
Qwen2.5-3B-Instruct	0.61	0.55	+0.06	3.1%
Qwen3.5-2B	0.62	0.48	+0.14	0.4%
Qwen3.5-4B	0.86	0.64	+0.22	0.4%
Qwen3.5-9B	0.86	0.79	+0.07	0.4%
Gemma-3-1b-it	0.55	0.02	+0.53	3.1%
Gemma-3-4b-it	0.97	0.68	+0.29	0.8%

Table 1: FVQ and CVQ at the smallest cell per block where every model has gap $> +0.05$ (K=20, N=50 proprietary; K=5, N=30 open-weight). **% Ctx** is the prompt's share of the model's advertised context window. No model uses more than **3.3%** of its window at the cell it fails on.

4 Pretraining Fails at In-Context Update Retrieval

Every model we evaluate exhibits a substantial FVQ–CVQ gap. At the representative cells in Table 1, gaps range from +6 to +53 percentage points, with the majority exceeding +20 pp. Per-cell evidence across the full (K, N) grid for all 20 models is in Appendix D. Figure 2 shows the trajectory in two open-weight models under three axis scans, including one that reaches the reversal regime.

The gap takes two shapes (Fig. 2): **primacy** (FVQ exceeds CVQ; all cells of Table 1) and **reversal** (CVQ exceeds FVQ but both well below ceiling, CVQ in the 50–65% range; smaller Qwen models at high load). Neither regime equals correct retrieval.

Intermediate-position queries ($1 < k < N$) collapse uniformly: at K=10, N=50 under the default Plain format, accuracy drops below 5% within two positions of the stream start across all five open-weight models and three of six proprietary models tested (Appendix E). Gemini-2.5-Pro is the only model at near-ceiling on this cell — but even it exhibits a +0.13 FVQ–CVQ gap at the harder K=20, N=50 cell (Table 1). The failure is universal; what differs is the operating point at which it becomes visible.

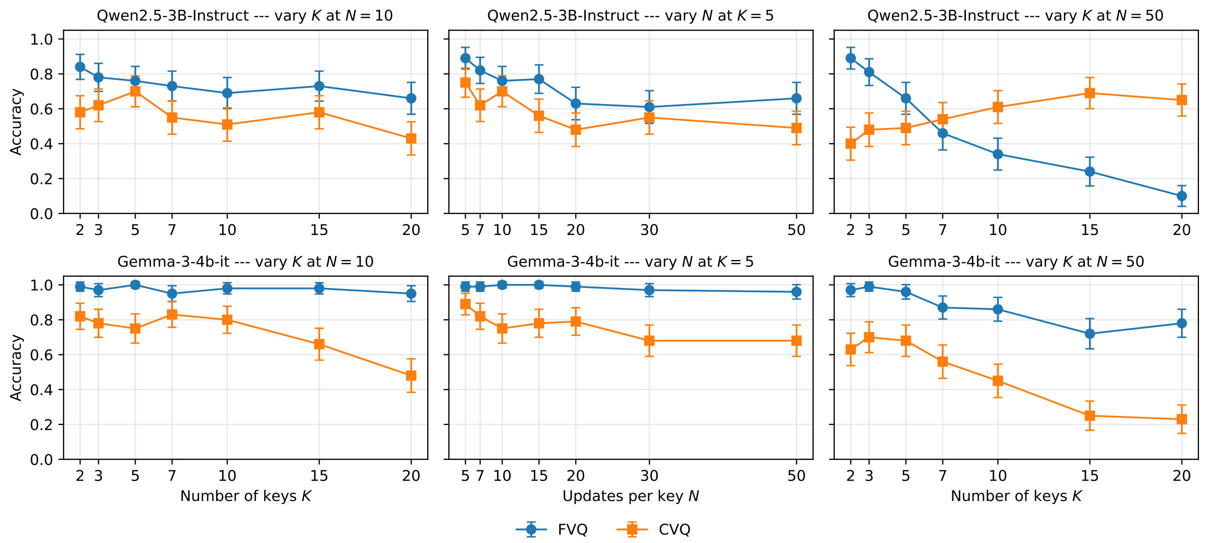

Figure 2: Qwen2.5-3B-Instruct (top) and Gemma-3-4b-it (bottom) across three axis scans on Semantic-Multi. Qwen crosses into the reversal regime at $K \approx 7$ (right column); Gemma stays in primacy throughout.

5 Two Recoveries: Format and LoRA

5.1 Format intervention

If the failure is about *locating* the relevant entries in the stream, making positions explicit at the surface should make the lookup easier. We test four formats: **Plain** (default, no markers); **Labeled** (per-entry index); **Block** (round headers); **Landmark** (round separators). Worked examples in App. B. Table 2 reports K=10, N=50 — the hardest cell in the format sweep — across 11 models (5 open-weight, 6 proprietary).

Block format eliminates the FVQ–CVQ gap across every open-weight model where we measure it (Table 2). The recovery is genuine, not equalisation: under Block both FVQ and CVQ sit near ceiling for every open-weight model where we have the data, whereas under Plain CVQ collapses while FVQ stays at ceiling. Proprietary models follow the same pattern at smaller magnitudes; Gemini-2.5-Pro is the only model at ceiling on both queries under Plain, consistent with its non-collapse in §4.

Labeled and Landmark produce mixed recoveries (Table 2; per-model breakdown + 2×2 decomposition in App. E): round-grouping and numbering each contribute to CVQ recovery, but no single property alone restores both queries.

5.2 LoRA intervention

We fit a LoRA adapter (Hu et al., 2022) to Qwen2.5-3B-Instruct on 18,000 examples from a small training grid ($K \leq 10$, $N \leq 20$). The adapter adds 7.4M trainable parameters — 0.24% of model weights. We evaluate on a held-out grid extending to K=30, N=75, on held-out categories, and cross-distribution on Semantic-Multi (full configuration in Appendix G).

On the 28 held-out Arbitrary-Single cells, both FVQ and CVQ accuracy reach 93–100% post-adapter (Table 3), roughly 10× beyond the training grid in stream length. The same adapter transfers to Semantic-Multi without modification (Table 4): at four held-out SEM cells, both queries reach 100% — including cells in the deep reversal regime with baseline gap up to −0.45. The dense intermediate-position queries that collapsed under the base model in §4 also recover to near-ceiling at every queried position across six cells (Figure 3; Appendix F); simultaneous recovery across positions, not just the last, rules out a pure recency-amplification account.

Family control. The same recipe applied to Gemma-3-4b-it — a different model family with the opposite baseline regime ($\leq +87$ pp primacy vs Qwen's reversal) — closes the gap on 28/28 held-out ARB cells (mean FVQ 99.4%, CVQ 97.0%) and 16/16 SEM OOD cells (mean FVQ 100%, CVQ 96.8%; Appendix G.1). The recovery is neither Qwen-specific nor regime-specific.

Negative control. The same recipe trained on GSM8K with identical hyperparameters learns arithmetic (16% → 70% GSM8K accuracy) but leaves the FVQ–CVQ gap unchanged on every held-out cell (Appendix G): the recovery is task-specific.

Model	Plain	Labeled	Block	Landmark
Open-weight models				
Qwen2.5-3B-Instruct	0.49 / 0.23	0.34 / 0.39	0.84 / 0.92	0.71 / 0.27
Qwen3.5-2B	0.95 / 0.01	0.85 / 0.93	1.00 / 0.98	0.80 / 0.07
Qwen3.5-4B	0.99 / 0.00	0.93 / 0.97	1.00 / 1.00	0.99 / 0.47
Gemma-3-4b-it	0.92 / 0.01	0.39 / 0.47	0.99 / 0.89	0.92 / 0.00
Proprietary models				
GPT-4.1	1.00 / 0.69	1.00 / 0.96	1.00 / 0.98	0.99 / 0.99
GPT-4.1-mini	1.00 / 0.77	1.00 / 0.94	0.99 / 0.99	0.78 / 1.00
Claude-4.5-Haiku	1.00 / 0.43	1.00 / 0.95	1.00 / 0.99	0.91 / 0.90
Claude-4.5-Sonnet	0.99 / 0.80	1.00 / 0.81	1.00 / 1.00	0.96 / 0.99
Gemini-2.5-Flash	1.00 / 0.76	1.00 / 0.98	1.00 / 1.00	1.00 / 0.95
Gemini-2.5-Pro	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	0.54 / 0.99

Table 2: Format-intervention results on Semantic-Multi ($K{=}10, N{=}50$). Each cell shows **FVQ / CVQ accuracy**; gap Δ = FVQ − CVQ. Plain is the default condition from §4. Wilson half-widths in Appendix E, Table 8.

Format and LoRA produce the same behavioural outcome via *operationally distinct internal loci*: pooled $\Pr(\text{attend to } v_{\text{last}})$ on the 15 LoRA-promoter heads is 0.16 under Block versus 0.68 under LoRA, with 0 of 15 heads reaching LoRA-level activation (App. J). Both restore intermediate-position retrieval (App. E, Figure 9; Figure 3). Together they demonstrate the capability is *latent*: present in the pretrained architecture but not surfaced by default.[1] §6 asks when in training this surfacing failure develops; §7 asks what changes mechanistically when LoRA recovers it.

6 When Does the Failure Develop?

We ask whether emergence is hiding at any training stage we have not yet evaluated. We track the FVQ–CVQ gap across 32 published SmolLM3-3B pretraining checkpoints (Hugging Face Smol Models Research Team, 2025) across all three documented stages, plus its five post-training stages (mid-training, SFT, APO, LC-expert, final). We evaluate on the 32-cell Arbitrary-Single grid partitioned by stream-load (**low**: $K{\leq}3$; **medium**: $K{\geq}5, N{<}20$; **high**: $K{\geq}5, N{\geq}20$); SmolLM2-1.7B (Ben Allal et al., 2025) provides a second family (App. H).

Figure 4 answers the question: no. The gap is directionally biased at every checkpoint in every regime; no regime stabilises at zero. The five post-training stages do not close it; SmolLM2 shows the same picture across its four-stage curriculum.

We evaluate in completion format for cross-checkpoint comparability — the earliest pretraining checkpoints predate chat-template training. Rerunning SmolLM3's post-training stages under the model's released chat template preserves the direction and ordering of the gap at every stage (Appendix H, Table 21); the failure is not a completion-format artifact.

The same checkpoints are documented to succeed at multi-step reasoning, code synthesis, and long-context retrieval at scales that exceed this task (Hugging Face Smol Models Research Team, 2025; Qwen Team, 2025). The capability is recoverable by a small adapter (§5.2); pretraining does not surface it at any checkpoint we measure, motivating the circuit-level explanation in §7.

7 Mechanism: What LoRA Changes

The recovery in §5.2 shows the capability is in the pretrained architecture; §6 shows it remains unsurfaced through every documented training stage. We ask what changes inside the model when LoRA surfaces it. All measurements use Qwen2.5-3B-Instruct on Arbitrary-Single at $K{=}2, N{=}5$ unless noted (layer indexing 0-based, L0–L35; Table 5 for bootstrap 95% CIs, Figure 5 for per-layer trajectories and spatial localisation).

7.1 The substrate already exists in baseline

Three linear probes characterise what the pretrained model already represents. A condition-discrimination probe (FVQ vs CVQ prompt) reaches 100% accuracy from layer 6 onward in *both* models; LoRA adds nothing. A FVQ-correctness probe reaches 81% at L33 in baseline — the model internally tracks its FVQ success. But a CVQ-correctness probe at L27–L33 sits near chance in baseline (57–65%), while the matching FVQ probe

[1] We treat a capability as latent when (i) a small, task-matched fine-tune restores it and (ii) the mechanism amplifies pre-existing structure rather than installing it from scratch. The arithmetic LoRA control (App G) fails (i); a model whose L30–L33 promoter heads had zero pre-LoRA $\Pr(\text{attend to } v_{\text{last}})$ would fail (ii) and falsify the claim for this task.

(K, N)	**Base gap**	**+ LoRA gap**
$(7, 30)$	$+12\%$	$+0\%$
$(7, 50)$	$+6\%$	$+4\%$
$(10, 30)$	-5%	$+0\%$
$(10, 50)$	-8%	$+0\%$
$(15, 20)$	-26%	$+2\%$
$(15, 30)$	-25%	$+2\%$
$(15, 50)$	-41%	$+0\%$
$(20, 30)$	-30%	$+2\%$
$(20, 50)$	-54%	$+4\%$
$(20, 75)$	-50%	$+4\%$
$(25, 30)$	-44%	$+2\%$
$(25, 50)$	-38%	$+2\%$
$(25, 75)$	-54%	$+7\%$
$(30, 30)$	-48%	$+2\%$
$(30, 50)$	-50%	$+6\%$
$(30, 75)$	-55%	$+6\%$

Table 3: Base vs +LoRA gap, 16 held-out cells extending to $K{=}30, N{=}75$ (adapter trained on $K \le 10, N \le 20$). Both queries reach 93–100 % post-adapter; FVQ/CVQ breakdown in Table 15, App. G.

	Base			**+ LoRA (ARB-trained)**		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
$(7, 20)$	60%	55%	+5%	100%	100%	+0%
$(15, 20)$	25%	55%	-30%	100%	100%	+0%
$(15, 30)$	35%	80%	-45%	100%	100%	+0%
$(20, 30)$	25%	65%	-40%	100%	100%	+0%

Table 4: Same LoRA, evaluated on Semantic-Multi (multi-token semantic values, never seen in training). All four held-out cells go from baseline reversal regime to gap 0%.

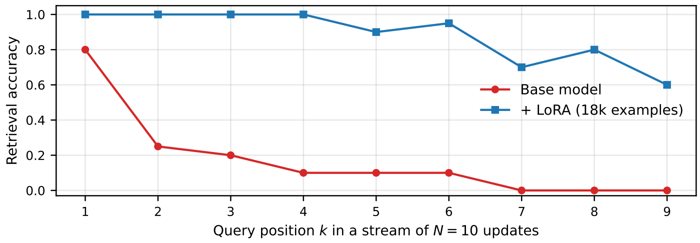

Figure 3: Per-position retrieval at $K{=}5, N{=}10$ on Arbitrary-Single, positions $k \in [1, N{-}1]$. Base collapses on intermediate positions; LoRA holds near ceiling at every position queried.

is at 77–81%: the baseline does not internally represent whether it will get the CVQ right.

Logit lens shows the parallel pattern at the token level. At L32 of the baseline, $\Pr(v_{\text{last}}){=}0.34$ — the answer surfaces briefly — then decays to 0.25 at L35. *Found then suppressed.* At a harder cell ($K{=}2, N{=}50$) the answer barely surfaces (0.02 at L32, 0.11 at L35), tracking the behavioural failure. The readout machinery in layers 34–35 is intact (attention patterns nearly identical pre/post-LoRA, Figure 5b); what the model lacks is propagation from the partial L32 answer to the final layer.

7.2 LoRA amplifies a partial v_{last} promoter circuit in layers 30–33

Attention routing locates the change. Fifteen heads in the L30–L33 band gain substantial v_{last} attention: pre-LoRA values of 0.05–0.23 rise to 0.60–0.86 post-LoRA — a 4–12× increase per head (Figure 5b). The largest shifts are L32H3 ($0.11 \rightarrow 0.78$), L32H7 ($0.13 \rightarrow 0.78$), L31H12 ($0.17 \rightarrow 0.81$), and L31H15 ($0.23 \rightarrow 0.86$). Layers 0–20 and 34–35 are essentially unchanged; the rewiring is spatially confined to a four-layer band.

The pre-LoRA values are not noise: the 15 promoters all sit above the 95th percentile of $\Pr(\text{attend to } v_{\text{last}})$ across all 576 baseline heads (median 0.001; only 2.3% of heads exceed 0.05). §7.3 confirms causally that the heads carry v_{last} signal pre-training: ablating the eight highest-attribution heads in the baseline drops L32 $\Pr(v_{\text{last}})$ by -0.28 (CI excludes zero).

The token-level consequence is immediate: post-LoRA L32 has $\Pr(v_{\text{last}}){=}0.99$ at $K{=}2, N{=}5$ and 0.62 at $K{=}2, N{=}50$, saturating to ~ 1.0 by L33–L35 at both. The representational consequence is parallel: the CVQ-correctness probe at L27–L33 rises from near-chance to 84–94% ($\Delta{=}{+}0.36$, 95% CI $[+0.21, +0.50]$ at L33), installed in the same late layers that already carried the FVQ-correctness signal.

LoRA does not act by removing suppression: ablating the five baseline suppressor heads (L26H3, L27H3, L29H3, L29H4, L30H3) in the LoRA model releases v_{last} at L31 *more* than in baseline ($\Delta{=}{+}0.150$ vs $+0.078$). The suppressors keep firing post-LoRA; LoRA strengthens a parallel promoter pathway that overpowers them.

7.3 Two LoRA changes: amplification and downstream redundancy

To distinguish what LoRA *installs* from what it *amplifies*, we ablate the eight highest-attribution promoter heads in both models with paired trial seeds; the within-trial Δs are directly comparable (Table 6).

Two findings emerge. First, the heads are partial pre-training carriers, and LoRA amplifies them: ablation drops L32 $\Pr(v_{\text{last}})$ by -0.28 in baseline (95% CI $[-0.34, -0.22]$, $n{=}200$) and by -0.40 in LoRA (95% CI $[-0.46, -0.33]$). The paired difference is $+0.12$ (95% CI $[+0.02, +0.21]$, excludes

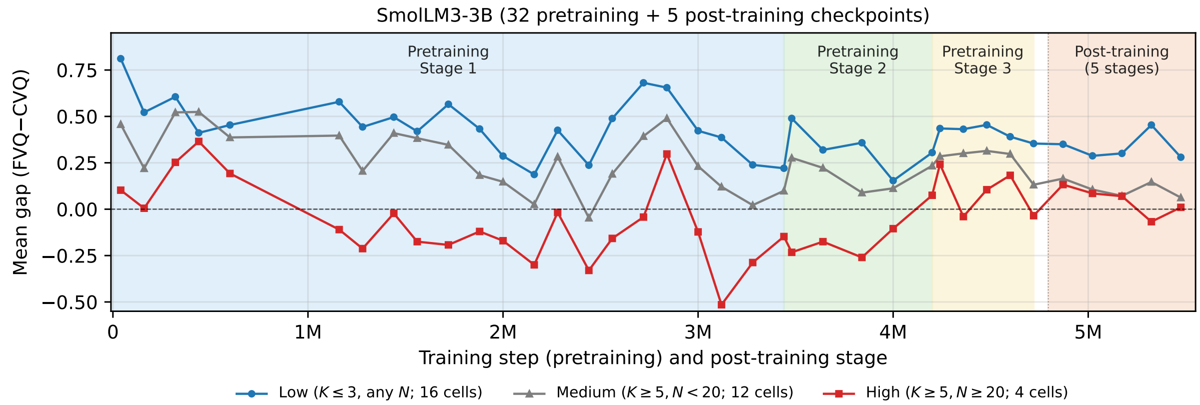

Figure 4: Mean FVQ−CVQ gap on Arbitrary-Single across the SmolLM3-3B (Hugging Face Smol Models Research Team, 2025) training trajectory. Three load regimes: **Low** ($K\leq3$, blue), **Medium** ($K\geq5, N<20$, grey), **High** ($K\geq5, N\geq20$, red). Pink band on the right: five post-training stages (mid-training, SFT, APO, LC-expert, final). No regime closes the gap at any documented checkpoint; SmolLM2-1.7B shows the same pattern (App. H).

zero) — post-LoRA the same eight heads carry $\sim 40\%$ more causal effect at L32.

Second, propagation to the final layer differs. In baseline, the ablation leaves a small detectable residue at L35 ($\Delta{=}-0.05$, 95% CI $[-0.10, -0.00]$). In LoRA, the same ablation is absorbed by L35 ($\Delta{=}+0.00$, CI $[0, 0]$); paired difference -0.05 (CI $[-0.10, -0.00]$, excludes zero). LoRA has installed compensatory pathways at L33+ that re-stabilise $\Pr(v_{\text{last}})$ when the upstream signal is partially knocked out — redundancy that does not pre-exist in baseline.

Cross-family replication. Probing, logit-lens, and attention-routing signatures replicate on Gemma-3-4b-it (different family, opposite baseline regime). The CVQ-correctness probe at L26–L33 rises from 0.70–0.73 to 0.82–0.89 (Δ between $+0.11$ and $+0.19$, all eight layers with CI excluding zero); logit lens at L33 shows $\Pr(v_{\text{last}})$ rising from 0.50–0.80 to ≈ 1.00 across three cells; attention routing identifies a top-15 promoter cluster spread across L14–L32 (Δ $+0.33$ to $+0.47$) — same late-layer functional pattern as Qwen, with a distributed rather than concentrated topology (App. I). Paired causal ablation on Gemma is left to future work.

Synthesis. The four methods form a closed causal chain: heads in L30–L33 attend to v_{last} (routing) $\rightarrow$ v_{last} builds at L32 (logit lens) $\rightarrow$ a CVQ-correctness feature becomes linearly decodable (probing) $\rightarrow$ behaviour recovers. LoRA does two things in this chain: it amplifies a pre-existing partial promoter circuit (4–12$\times$ in 15 heads of L30–L33) and installs compensatory pathways at L33+ that absorb upstream perturbations. The pretraining failure is a gain and robustness failure, not a representational one — §8 discusses what this implies.

8 Discussion

When each recovery is usable. The two recoveries operate under different deployment constraints. Format intervention works at inference time but requires the deployer to control the prompt — system-prompted retrieval pipelines, agent harnesses with structured state, or instrumented multi-turn dialogue. It does not apply when prompt structure is determined by user input or upstream components the deployer does not control. LoRA requires weight access and so applies to open-weight or in-house fine-tuneable models, but not to proprietary frontier inference behind a query interface. For frontier-only deployment paths neither recovery is available; our results predict the current-value retrieval failure persists.

Implications for pretraining. The next-token loss provides no direct supervision for tracking which previous assignment to a slot is currently valid; the 79-checkpoint sweep (§6) is consistent with this. A training signal that directly rewards in-context update tracking is a natural next step; our LoRA is one proof-of-concept.

Limitations

Real-world stateful benchmarks. Our task is a controlled synthetic instance; linking the gap to application-level failures (multi-turn dialogue degradation, Laban et al., 2025; entity state-tracking, Rezaee et al., 2025) is open.

Method	Quantity	Baseline	+LoRA	Δ [95% CI]
Probe	CVQ-correctness, L33	0.58	0.94	+0.36 [+0.21, +0.50]
Probe	CVQ-correctness, L32	0.65	0.94	+0.29 [+0.22, +0.36]
Probe	CVQ-correctness, L30	0.57	0.84	+0.28 [+0.19, +0.37]
Probe	FVQ-correctness, L33	0.81	0.93	+0.12 [+0.05, +0.19]
Logit lens	$\Pr(v_{\text{last}})$, L32, K=2, N=5	0.34	0.99	+0.65 [+0.55, +0.73]
Logit lens	$\Pr(v_{\text{last}})$, L32, K=2, N=10	0.21	0.93	+0.73 [+0.63, +0.81]
Logit lens	$\Pr(v_{\text{last}})$, L32, K=2, N=50	0.02	0.62	+0.60 [+0.50, +0.69]
Logit lens	$\Pr(v_{\text{last}})$, L35, K=2, N=50	0.11	0.90	+0.79 [+0.71, +0.87]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L32H3	0.11	0.78	+0.68 [+0.65, +0.71]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L31H12	0.16	0.81	+0.66 [+0.60, +0.71]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L32H7	0.12	0.78	+0.65 [+0.62, +0.68]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L31H15	0.23	0.86	+0.64 [+0.57, +0.70]

Table 5: Bootstrap 95% CIs on key mechanistic Δs at L30–L33, baseline vs +LoRA. Probing and attention-routing rows aggregate within the band; logit-lens rows give $\Pr(v_{\text{last}})$ at L32. Trial counts: probing 200 trials at K=2, N=5; logit lens 100 trials per cell; attention routing 50 trials at K=2, N=30. Bootstrap n_{boot}=2000, seed 17.

Ablation condition	Δ at L31	Δ at L32	Δ at L33	Δ at L35
Baseline + 5 baseline-suppressor heads	+0.078	−0.009	−0.058	+0.015
+LoRA + 5 baseline-suppressor heads	+0.150	−0.001	+0.000	+0.000
+LoRA + 8 LoRA-promoter heads	+0.007	**−0.396**	−0.005	+0.000
Baseline + 8 LoRA-promoter heads (control)	+0.014	**−0.277**	−0.007	−0.051

Table 6: Stage 3C four-way causal-ablation summary. Each row: paired within-run Δ= ablated − normal at four layers (K=2, N=5, CVQ; promoter rows n=200, suppressor rows n=50). **Bold row** is the baseline control — the eight LoRA-promoter heads carry substantial L32 effect already in baseline, establishing them as partial v_{last} carriers pre-LoRA. The L33–L35 columns show LoRA's new downstream redundancy: baseline leaks to the readout, LoRA absorbs it.

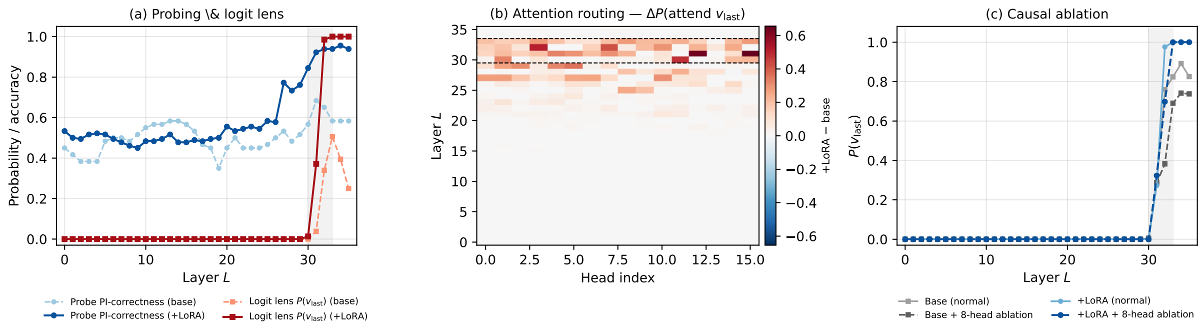

Figure 5: Mechanism summary for Qwen2.5-3B-Instruct (K=2, N=5). **(a)** Per-layer $\Pr(v_{\text{last}})$ (logit lens) and CVQ-correctness probe accuracy, baseline vs +LoRA — both signals converge on the L27–L33 band (shaded). **(b)** Per-head Δ in $\Pr(\text{attend to } v_{\text{last}})$ across all 576 heads; the bright band at L30–L33 (dashed) contains the 15 promoter heads. **(c)** Stage 3C ablation of the eight top-attribution promoter heads drops $\Pr(v_{\text{last}})$ at L32 in *both* baseline and LoRA models; baseline ablation propagates to the readout, LoRA absorbs by L33+.

Scale and language coverage. English only; the trajectory analysis is bounded by the public checkpoint releases ($\leq$ 3B). Extending to $\geq$ 7B, non-Transformer architectures, other languages, and format families beyond our four is future work.

LoRA scope. Our main LoRA is attention-only ($\{Q, K, V, O\}$, rank 16). A 5-cell preview with MLP-only LoRA also closes the gap (App. K); full-grid evaluation and mechanism characterisation of the MLP pathway are open.

Open questions. A prose-narrative variant built on Dota 2 match commentary could not be cleanly constructed (position-indexed retrieval is confounded with natural event ordering). And what model property predicts the primacy-vs-reversal split — Qwen-family crosses into reversal, Gemma-family does not — is open.

Ethical Considerations

Experiments use synthetic key–value streams from public English vocabulary; no human subjects, no

PII. Open-weight models are used under their licenses; proprietary models via APIs under standard terms of service. Our findings inform responsible deployment: LLMs should not be relied upon to track repeatedly-updated in-context variables without explicit positional cues or targeted fine-tuning.

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A Dataset Details

We construct two datasets that share the same (K, N) schema and grid but differ in surface form: **Arbitrary-Single (ARB)** uses single-token English words as values, while **Semantic-Multi (SEM)** uses multi-token real category members. Both isolate position-indexed retrieval from semantic shortcuts.

Arbitrary-Single (ARB). A pool of 2,300 single-token English words organised into 46 semantic categories (e.g., *color*, *fruit*, *metal*). For each trial, K categories are sampled as keys and N values are drawn without replacement from each category's pool. Keys and values share no inherent update semantics — there is no reason to expect any particular ordering among values of the same category. This isolates tracking from semantic shortcuts and provides the single-token values required for the mechanistic analysis (§7).

Semantic-Multi (SEM). A pool of 2,403 multi-token values across the same 46 categories, where values are real members of each category (e.g., *gemstone* $\rightarrow$ *alexandrite*, *amethyst*, …). Values are multi-token and drawn from a smaller per-category pool (45–70 values), limiting feasible update counts to $N \leq 50$ for most key counts. SEM is the primary behavioural dataset because the realistic key–value semantics rule out trivial template-matching strategies.

Grid coverage. On the unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$ (§3), cells where the prompt exceeds a model's context limit are excluded; cells where the dataset pool is insufficient are also excluded (relevant for SEM at high N). This yields up to 79 feasible cells per model on ARB and up to 63 on SEM.

B Prompts, Matching Rule, and Seeds

This appendix documents the exact text used to construct the inference prompts, the four format variants used in §5, the string-matching rule used to score model responses, and the deterministic seeding scheme.

Naming convention. The main text refers to the second query type as *CVQ* (current-value query), reflecting its meaning. The actual prompt text asks for the *last* value of the target key, which is operationally equivalent when the stream is the complete record of updates.

System prompt and queries

System prompt:
You are a precise data extraction tool. Output ONLY a single word — the exact value requested. No other text, no explanation, no punctuation.

Queries (rendered with the target key substituted in):
FVQ: *What was the first value of* [TARGET]*?*
CVQ: *What was the last value of* [TARGET]*?*
IVQ: *What was the* [K]*-th value of* [TARGET]*?*

Figure 6: Inference-time system prompt and per-query phrasing used in all behavioural sweeps reported in the paper. "CVQ" in the main text corresponds to the literal "last value of" query above.

Stream formats. The four formats compared in §5 differ only in how the stream lines are rendered and which user-turn preamble introduces them. Figure 7 shows a small (K=2, N=3) example in each format.

Chat-template rendering. For instruction-tuned models the user/system messages are rendered through the model's tokenizer-default chat template; for base models without a chat template we use a completion-style few-shot wrapper described in the released code.

Plain (`flat_nolabel`; default in §4)

Read the following key-value stream. Each key appears multiple times as it gets updated. Count each occurrence of a key as one update.

```
color: red
size: large
color: blue
size: small
color: green
size: medium
```

What was the 3rd value of color?
Answer with ONLY the exact value. No explanation.

Labeled (`flat_verbose`)

Read the following key-value stream. Each key is updated multiple times.

```
color (update 1): red
size (update 1): large
color (update 2): blue
size (update 2): small
color (update 3): green
size (update 3): medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Block (`block`)

Read the following key-value stream. Each key is updated multiple times, grouped by update round.

```
[Update 1]
color: red
size: large
[Update 2]
color: blue
size: small
[Update 3]
color: green
size: medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Landmark (`landmark`)

Read the following key-value stream. Each key is updated multiple times, grouped by update round. The '—' marker separates consecutive update rounds.

```
color: red
size: large
–
color: blue
size: small
–
color: green
size: medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Figure 7: The four prompt formats compared in §5, illustrated on a $K{=}2, N{=}3$ instance. Same underlying stream content; only the surface structure changes. The "Plain" format is the default for all main-body figures and tables (§4).

Matching rule. A model response is counted as correct if, after lowercasing and stripping whitespace, the expected value either equals the response, is contained in the response, or is a prefix of the response. This prefix-tolerant case-insensitive rule accommodates multi-token semantic values where models occasionally emit a trailing qualifier; spot-checks confirm it does not credit lexically unrelated strings.

Seeds. For each $(K, N, \text{query type}, \text{trial index})$ tuple we derive a deterministic seed by hashing the tuple. This ensures the identical stream is presented across base-vs-intervention comparisons at the same cell and trial index, so the two conditions are scored on identical inputs.

C Model Details

Table 7 lists every model evaluated in this paper. The first block is the main open-weight set whose behaviour is analysed in the main text; the second block is three smaller auxiliary models reported in Appendix D for completeness; the third block is the proprietary frontier models. IT = instruction-tuned.

Model licenses. All open-weight models are used under the license listed on each model's HuggingFace page (verified individually). Proprietary models are accessed through their official APIs under each provider's terms of service.

- `Qwen/Qwen2.5-0.5B-Instruct` (HuggingFace): **Apache 2.0**.
- `Qwen/Qwen2.5-1.5B-Instruct` (HuggingFace): **Apache 2.0**.
- `Qwen/Qwen2.5-3B-Instruct` (HuggingFace): **Qwen Research License** (`https://huggingface.co/Qwen/Qwen2.5-3B-Instruct/blob/main/LICENSE`).
- `Qwen/Qwen2.5-3B` (HuggingFace): **Qwen Research License**.
- `Qwen/Qwen3.5-0.8B` (HuggingFace): **Apache 2.0**.
- `Qwen/Qwen3.5-2B` (HuggingFace): **Apache 2.0**.

Model	**Family**	**Params**	**IT?**	**Ctx**
Main open-weight				
Qwen2.5-0.5B-Instruct	Qwen2.5	0.5B	✓	32K
Qwen2.5-1.5B-Instruct	Qwen2.5	1.5B	✓	32K
Qwen2.5-3B-Instruct	Qwen2.5	3B	✓	32K
Qwen2.5-3B	Qwen2.5	3B	×	32K
Qwen3.5-0.8B	Qwen3.5	0.8B	✓	256K
Qwen3.5-2B	Qwen3.5	2B	✓	256K
Qwen3.5-4B	Qwen3.5	4B	✓	256K
Qwen3.5-9B	Qwen3.5	9B	✓	256K
Gemma-3-270m-it	Gemma-3	270M	✓	32K
Gemma-3-1b-it	Gemma-3	1B	✓	32K
Gemma-3-4b-it	Gemma-3	4B	✓	128K
Auxiliary (appendix only)				
TinyLlama-1.1B	LLaMA	1.1B	✓	2K
StableLM-2-1.6B	StableLM	1.6B	✓	4K
Pythia-410M	Pythia	410M	×	2K
Proprietary frontier				
GPT-4.1	OpenAI	—	—	1M
GPT-4.1-mini	OpenAI	—	—	1M
Claude-4.5-Haiku	Anthropic	—	—	200K
Claude-4.5-Sonnet	Anthropic	—	—	1M
Gemini-2.5-Flash	Google	—	—	1M
Gemini-2.5-Pro	Google	—	—	1M

Table 7: All 20 models evaluated, grouped by role: 11 main open-weight models that drive the main analyses; 3 smaller auxiliary models reported only in Appendix D; and 6 proprietary frontier models. **Ctx** is the advertised input context window (HF `max_position_embeddings` for open-weight; provider documentation for proprietary; see Appendix D).

- `Qwen/Qwen3.5-4B` (HuggingFace): **Apache 2.0**.
- `Qwen/Qwen3.5-9B` (HuggingFace): **Apache 2.0**.
- `google/gemma-3-270m-it` (HuggingFace): **Gemma Terms of Use** (https://ai.google.dev/gemma/terms).
- `google/gemma-3-1b-it` (HuggingFace): **Gemma Terms of Use**.
- `google/gemma-3-4b-it` (HuggingFace): **Gemma Terms of Use**.
- `TinyLlama/TinyLlama-1.1B-Chat-v1.0` (HuggingFace): **Apache 2.0**.
- `stabilityai/stablelm-2-1_6b-chat` (HuggingFace): **Stability AI Non-Commercial Research Community License** (https://huggingface.co/stabilityai/stablelm-2-1_6b-chat/blob/main/LICENSE.md).
- `EleutherAI/pythia-410m` (HuggingFace): **Apache 2.0**.
- `HuggingFaceTB/SmolLM2-1.7B-Instruct` (HuggingFace): **Apache 2.0**.
- `HuggingFaceTB/SmolLM3-3B` (HuggingFace): **Apache 2.0**.
- **GPT-4.1, GPT-4.1-mini** (OpenAI API): OpenAI Terms of Service (https://openai.com/policies/terms-of-use/).
- **Claude-4.5-Haiku, Claude-4.5-Sonnet** (Anthropic API): Anthropic Commercial Terms of Service (https://www.anthropic.com/legal/commercial-terms).
- **Gemini-2.5-Flash, Gemini-2.5-Pro** (Google API): Gemini API Additional Terms of Service (https://ai.google.dev/gemini-api/terms).

The LoRA adapter we release is distributed under the **Qwen Research License**, matching the license of its base model (`Qwen/Qwen2.5-3B-Instruct`).

D Full Model Results

Context-usage measurement (Table 1). The **% Ctx** column of Table 1 reports the fraction of each model's advertised context window used by one canonical Semantic-Multi prompt at the row's representative cell. Open-weight rows use the K=5, N=30 prompt (**1,029** tokens); proprietary rows use the K=20, N=50 prompt (**6,565** tokens). All counts use `tiktoken o200k_base` (the GPT-4o / GPT-4.1 tokeniser) for uniform cross-provider comparability; on-row tokenisation with each model's actual tokeniser changes individual counts by under $\pm 5\%$ for English text and does not move any row above 5% of its window.

Context-window sources. Open-weight windows are read from each model's `config.json` on the HuggingFace Hub (`max_position_embeddings` in `text_config` for multi-modal models): Qwen2.5-3B-Instruct 32,768; Qwen3.5-2B, -4B, -9B 262,144 each; Gemma-3-1b-it 32,768; Gemma-3-4b-it 131,072 (rope-scaling factor 8 over the 16,384-token base). Proprietary windows are advertised input limits from provider documentation: GPT-4.1 and GPT-4.1-mini 1,000,000 tokens (https://openai.com/index/gpt-4-1/); Claude-4.5-Haiku 200,000 tokens; Claude-4.5-Sonnet 1,000,000 tokens via the long-context option (https://docs.anthropic.com/en/docs/about-claude/models/overview); Gemini-2.5-Flash 1,048,576 and Gemini-2.5-Pro 1,000,000 (https://ai.google.dev/gemini-api/docs/models).

E Format Intervention Data

This appendix supports two claims made in the main text: the format-intervention results in §5.1 (Table 2, in §5.1) and the multi-model intermediate-position retrieval claim in §4.

Per-model patterns under Labeled and Landmark. The full 4-format $\times$ 11-model results in Table 2 (§5.1) reveal two distinct partial-recovery patterns for the intermediate formats.

Labeled (per-entry numbering, no grouping) tends to push FVQ and CVQ together rather than restoring CVQ to ceiling. In Qwen2.5-3B-Instruct, FVQ drops from 0.49 (Plain) to 0.34 while CVQ rises only to 0.39 — both land near 0.4, shifting the failure rather than removing it. Gemma-3-4b-it shows the same collapse: FVQ falls from 0.92 to 0.39 and CVQ stays near 0.47. The proprietary set tolerates Labeled better, suggesting that per-entry indexing is a noisy cue smaller open-weight models cannot disambiguate from update content.

Landmark (round separators, no numbers) preserves FVQ but fails to recover CVQ in the open-weight set: Gemma-3-4b-it stays at FVQ$\approx$0.97 / CVQ$\approx$0.00, Qwen3.5-2B at FVQ$\approx$0.80 / CVQ$\approx$0.07. Proprietary models recover more cleanly under Landmark, but the gap remains positive there.

Read together, Labeled and Landmark localise Block's two effective ingredients: round structure provides salient boundaries for the lookup region (without which CVQ stays floored), while explicit round numbers anchor that structure to the query (without which open-weight models fall back to Plain-format behaviour). Block combines both.

Wilson 95% half-widths. Table 8 reports the same per-(model, format) FVQ and CVQ accuracies as Table 2 with Wilson 95% half-widths (Wilson, 1927; up to 200 trials per cell). Half-widths are ≤ 0.07 throughout; the Plain-to-Block CVQ recoveries cited in §5.1 (e.g. Qwen3.5-2B $0.01 \rightarrow 0.98$, Qwen3.5-4B $0.00 \rightarrow 1.00$, Gemma-3-4b-it $0.01 \rightarrow 0.89$) exceed combined-CI overlap by a wide margin. The Labeled equalisation cases discussed above (Qwen2.5-3B-Instruct FVQ $0.49 \rightarrow 0.34$, CVQ $0.23 \rightarrow 0.39$; Gemma-3-4b-it FVQ $0.92 \rightarrow 0.39$, CVQ $0.01 \rightarrow 0.47$) show CIs that exclude the corresponding Plain values in every case.

Multi-model per-position retrieval under the default Plain format. Figure 8 shows per-position retrieval accuracy at $K{=}10, N{=}50$ on Arbitrary-Single under the default Plain format, separately for the five open-weight models in the ucurve sweep and the six proprietary frontier models. All five open-weight models collapse below 5% accuracy within two positions of the stream start. The proprietary panel reveals three regimes: three models (GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku) collapse like the open-weight set; Claude-4.5-Sonnet and Gemini-2.5-Flash show partial recovery; Gemini-2.5-Pro alone maintains near-ceiling accuracy across all queried positions.

Block format restores per-position retrieval. Figure 9 compares per-position accuracy under Plain vs Block format on the same five open-weight models. Plain collapses below 5% accuracy by position 2 across every model; Block restores accuracy to 0.70–0.99 at every queried position. This substantiates the §5.2 close claim that the format intervention recovers intermediate-position retrieval, not only the FVQ–CVQ gap.

Block 2×2 decomposition. The Block format combines two design properties: round-grouping (entries clustered by update round) and round-numbering (headers labelling each round). The

Model	Plain	Labeled	Block	Landmark
Open-weight models				
Qwen2.5-3B-Instruct	0.49 / 0.23	0.34 / 0.39	0.84 / 0.92	0.71 / 0.27
	±0.07 / ±0.06	±0.06 / ±0.07	±0.05 / ±0.04	±0.06 / ±0.06
Qwen3.5-2B	0.95 / 0.01	0.85 / 0.93	1.00 / 0.98	0.80 / 0.07
	±0.03 / ±0.02	±0.05 / ±0.04	±0.01 / ±0.02	±0.06 / ±0.03
Qwen3.5-4B	0.99 / 0.00	0.93 / 0.97	1.00 / 1.00	0.99 / 0.47
	±0.02 / ±0.01	±0.04 / ±0.02	±0.02 / ±0.02	±0.01 / ±0.07
Gemma-3-4b-it	0.92 / 0.01	0.39 / 0.47	0.99 / 0.89	0.92 / 0.00
	±0.04 / ±0.02	±0.07 / ±0.07	±0.02 / ±0.04	±0.04 / ±0.01
Proprietary models				
GPT-4.1	1.00 / 0.69	1.00 / 0.96	1.00 / 0.98	0.99 / 0.99
	±0.01 / ±0.06	±0.02 / ±0.05	±0.03 / ±0.04	±0.01 / ±0.01
GPT-4.1-mini	1.00 / 0.77	1.00 / 0.94	0.99 / 0.99	0.78 / 1.00
	±0.01 / ±0.06	±0.02 / ±0.05	±0.03 / ±0.03	±0.06 / ±0.01
Claude-4.5-Haiku	1.00 / 0.43	1.00 / 0.95	1.00 / 0.99	0.91 / 0.90
	±0.01 / ±0.07	±0.02 / ±0.04	±0.01 / ±0.02	±0.04 / ±0.04
Claude-4.5-Sonnet	0.99 / 0.80	1.00 / 0.81	1.00 / 1.00	0.96 / 0.99
	±0.01 / ±0.06	±0.01 / ±0.05	±0.04 / ±0.04	±0.03 / ±0.01
Gemini-2.5-Flash	1.00 / 0.76	1.00 / 0.98	1.00 / 1.00	1.00 / 0.95
	±0.01 / ±0.06	±0.03 / ±0.04	±0.03 / ±0.03	±0.02 / ±0.05
Gemini-2.5-Pro	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	0.54 / 0.99
	±0.03 / ±0.03	±0.04 / ±0.04	±0.04 / ±0.04	±0.07 / ±0.01

Table 8: Format-intervention results with Wilson 95% half-widths, $K=10, N=50$. Each cell shows FVQ / CVQ on the first line and the matching ±half-widths on the second. Same data as Table 2.

Labeled and Landmark formats separate these: Labeled uses per-entry numbering without grouping, Landmark uses grouping without numbering. We decompose Block's effect across 36 (model, K, N) cells from the proprietary ucurve sweep, bootstrap 95% CIs ($n_{\text{boot}}=2000$, seed 17).

CVQ contrast	**Δ [95% CI]**
Numbering (Labeled − Plain)	$+0.106$ [$+0.065$, $+0.153$]
Grouping (Landmark − Plain)	$+0.136$ [$+0.099$, $+0.177$]
Combined (Block − Plain)	$+0.143$ [$+0.085$, $+0.213$]
Super-additivity	-0.100 [-0.143, -0.062]
FVQ contrast	**Δ [95% CI]**
Numbering (Labeled − Plain)	$+0.002$ [-0.000, $+0.005$]
Grouping (Landmark − Plain)	-0.142 [-0.210, -0.086]
Combined (Block − Plain)	-0.001 [-0.005, $+0.001$]
Super-additivity	$+0.141$ [$+0.085$, $+0.213$]

Table 9: Block 2×2 decomposition. Both numbering and grouping contribute to CVQ recovery, but their combination undershoots the sum (sub-additive). For FVQ, grouping alone hurts (-14.2 pp) and numbering rescues; Block's no-cost FVQ behaviour depends on the combination.

Reading the decomposition. For **CVQ**, both ingredients help independently ($+10.6$ pp from numbering, $+13.6$ pp from grouping; both CIs exclude zero). Block's combined gain ($+14.3$ pp) is less than the sum (super-additivity -0.10, CI excludes zero): the two ingredients partially overlap. For **FVQ**, numbering alone is null and grouping alone *hurts* (-14.2 pp); Block recovers because numbering compensates the grouping cost. Block's ability to lift CVQ *without* sacrificing FVQ is therefore not attributable to either property alone — it depends on the combination.

F Dense Intermediate-Position Retrieval — Full Sweep

Figure 3 in §5 shows position-by-position retrieval accuracy at a single cell ($K=5, N=10$). To substantiate the broader claim that the failure is a general loss of position-indexed retrieval, we report the same measurement across all six cells we tested. Figure 10 shows the per-position curves; Table 10 summarises first / intermediate / last position accuracy per cell. Across all six (K, N) cells, baseline intermediate-position accuracy stays at or below 25% and the LoRA-tuned model recovers near-ceiling accuracy at every position queried.

G LoRA Training Details

This appendix documents the full configuration of the LoRA intervention reported in §5.2. All scripts, training data, and checkpoint selection logic are in the released codebase.

Adapter and optimiser. The adapter is fitted with PEFT (Hu et al., 2022) on top of the frozen base model. Table 11 lists the hyperparameters; LoRA matrices are inserted into the four attention projections $\{Q, K, V, O\}$ of every transformer

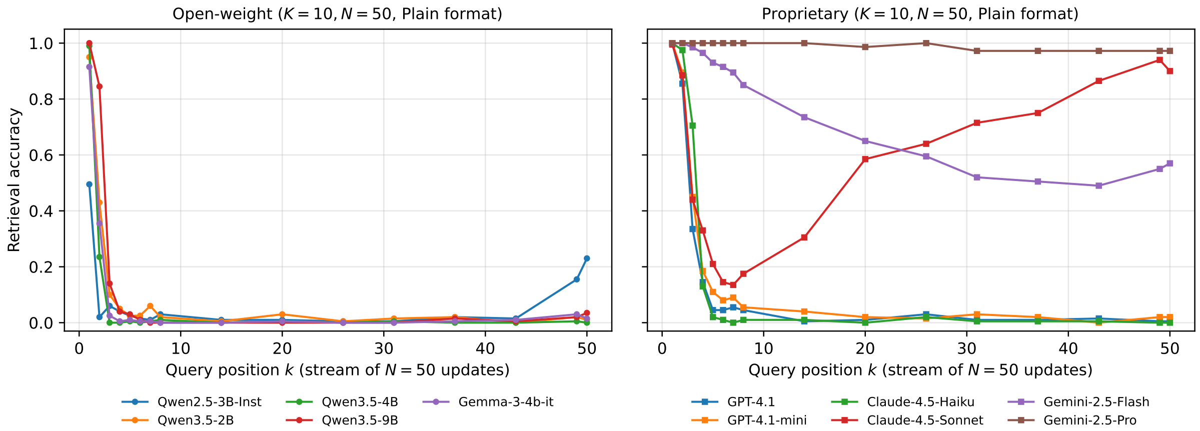

Figure 8: Per-position retrieval accuracy on Arbitrary-Single at $K{=}10, N{=}50$ under the default Plain format (ucurve sweep; `flat_nolabel` condition). Each line is one model. **Left**: five open-weight models, all collapsing uniformly to near-floor by position 2. **Right**: six proprietary models; most collapse but Gemini-2.5-Pro maintains near-ceiling accuracy throughout, and Claude-4.5-Sonnet shows U-shaped recovery toward the end of the stream.

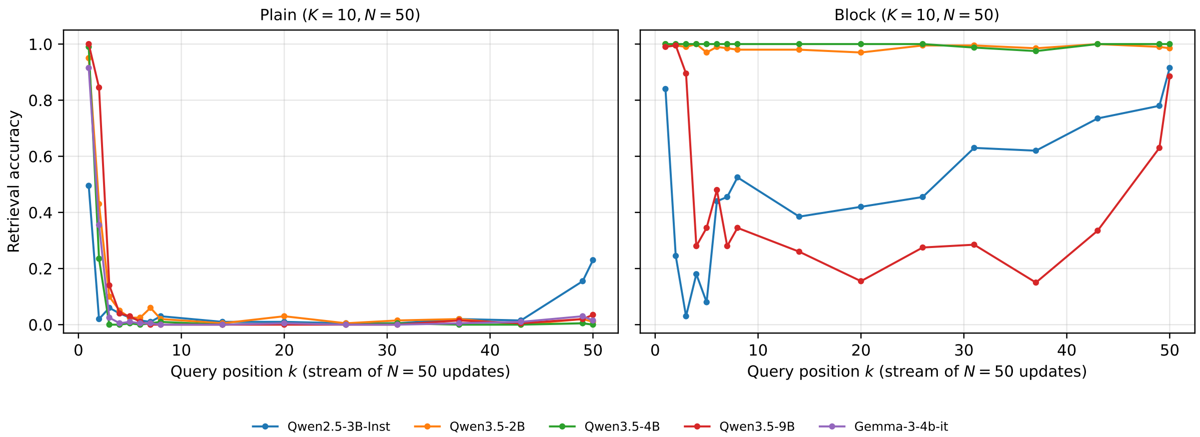

Figure 9: Per-position retrieval at $K{=}10, N{=}50$ on Arbitrary-Single, five open-weight models, Plain (left) vs Block (right) format. Block restores intermediate-position accuracy across every model, paralleling the LoRA recovery shown for Qwen2.5-3B-Instruct in Figure 3.

layer, and MLP and embedding weights are left unmodified.

Training data composition. Each training example is one (K, N) stream with a single query appended. The 18,000 training examples are split 40 % FVQ, 40 % CVQ, 20 % intermediate-position queries (IVQ). Including IVQ during training removes a possible shortcut: a model that simply amplified a recency-attending circuit could pass FVQ plus CVQ supervision while leaving intermediate positions unchanged; IVQ supervision blocks that path. Training examples are sampled uniformly across the 16 training cells, yielding $\sim$1,125 examples per cell.

Training and held-out cell split. The training and evaluation cells are disjoint by design. Table 12 lists both grids; held-out cells extend roughly 10$\times$ beyond training in stream length ($K \times N$ from ≤ 200 at training to $\leq 2{,}250$ at evaluation).

Category split. Of the 46 semantic categories in the Arbitrary-Single pool (Appendix A), 35 are used during training and 11 are held out for evaluation; no category appears in both splits (Table 13).

Compute Budget and Software Stack. Open-weight model evaluation (14-model behavioural sweep, format-intervention sweep, intermediate-position sweep, training-dynamics across 79 SmolLM2/3 checkpoints, mechanism probing) and the main Qwen2.5-3B-Instruct LoRA training ran on a single NVIDIA L40S GPU node, totalling roughly 120 GPU-hours wall-clock. The Gemma-3-4b-it family-control LoRA and the cross-family mechanism analyses ran on a SageMaker NVIDIA L4 node. Proprietary-model evaluation is via offi-

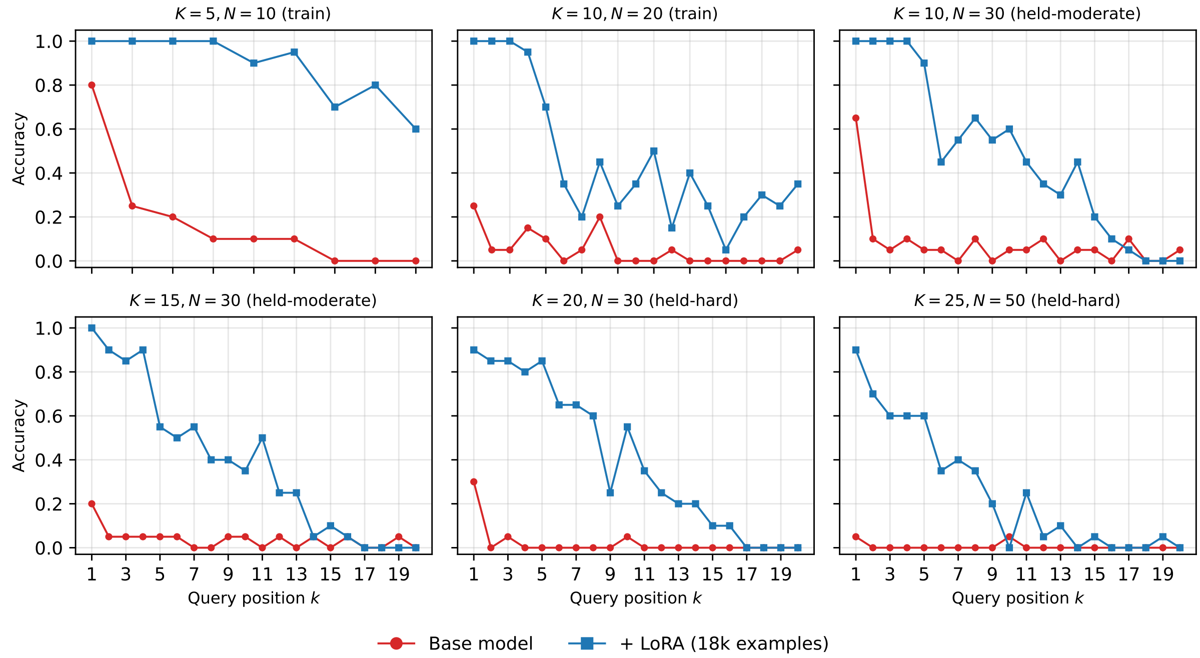

Figure 10: Per-position retrieval accuracy at all six tested (K, N) cells on Arbitrary-Single, base model vs LoRA-tuned model, for positions $k \in [1, N-1]$ ($k{=}N$ is omitted because the experiment script substitutes a CVQ "last value" query at that position). Group labels (`train`, `held-moderate`, `held-hard`) indicate whether the cell was included in the LoRA training grid.

	Base			**+ LoRA**		
(K, N)	Pos. 1	Interm. min–max	Last	Pos. 1	Interm. min–max	Last
$(5, 10)$	80%	0%–25%	0%	100%	70%–100%	60%
$(10, 20)$	25%	0%–20%	5%	100%	5%–100%	35%
$(10, 30)$	65%	0%–10%	5%	100%	0%–100%	0%
$(15, 30)$	20%	0%–5%	0%	100%	0%–90%	0%
$(20, 30)$	30%	0%–5%	0%	90%	0%–85%	0%
$(25, 50)$	5%	0%–5%	0%	90%	0%–70%	0%

Table 10: Per-cell summary of dense intermediate-position retrieval accuracy on Arbitrary-Single. "Interm. min–max" is the range across all intermediate positions ($1 < k < N$) at that cell. The base model loses accuracy at every intermediate position across every cell; the LoRA-tuned model holds near ceiling.

Setting	**Value**
Base model	Qwen2.5-3B-Instruct
LoRA rank	16
LoRA α	32
LoRA dropout	0.05
Target modules	q/k/v/o_proj
Trainable parameters	7.4M (0.24% of model)
Learning rate	2×10^{-4}
LR schedule	cosine, 100-step linear warm-up
Optimiser	paged AdamW (8-bit)
Epochs	2
Batch size	8 (×8 accum = eff. 64)
Max sequence length	2,048
Mixed precision	bfloat16
Training examples	18k (+2k val, different seeds)
Checkpoint selected	best val loss (step 564, end of ep. 2)

Table 11: LoRA training hyperparameters.

Split	**Cells** (K, N)
Training (16)	$K \in \{2, 3, 5, 10\}$, $N \in \{5, 10, 15, 20\}$
Held-out (28)	$K{=}7$, $N \in \{30, 50, 75\}$
	$K{=}10$, $N \in \{30, 50, 75\}$
	$K{=}15$, $N \in \{15, 20, 30, 50, 75\}$
	$K{=}20$, $N \in \{15, 20, 30, 50, 75\}$
	$K{=}25$, $N \in \{10, 15, 20, 30, 50, 75\}$
	$K{=}30$, $N \in \{10, 15, 20, 30, 50, 75\}$

Table 12: Training and held-out cells for the LoRA intervention. Training cells cover only low (K, N); held-out cells extend to $K{=}30$ and $N{=}75$, well outside the training distribution in both dimensions.

Training (35)	**Held-out (11)**
visual art, tools, landform, musical instrument, gemstone, fabric, tree species, cheese variety, architectural style, cloud formation, bird species, culinary herb, flower species, wine variety, dance style, pasta shape, literary genre, cooking method, mathematical concept, weather phenomenon, ocean current, mineral type, coffee variety, telescope type, martial art, sea creature, psychology term, chemical element, dinosaur genus, programming language, ancient civilization, bridge type, photography technique, boat type, cartoon character	*stadium name, surgical procedure, constellation, spice blend, guitar type, hat style, painting medium, volcano name, fruit variety, sword type, board game*

Table 13: LoRA category split. 35 categories used during training, 11 held out for evaluation; no overlap.

cial APIs and uses no local GPU. All decoding is greedy (temperature 0) with maximum new tokens 32.

Hardware:

- NVIDIA L40S (48 GB VRAM): main open-weight evaluation, Qwen2.5-3B-Instruct LoRA training. ∼120 GPU-hours total.
- NVIDIA L4 (23 GB VRAM, Ubuntu 24.04 LTS, driver 595.58.03): Gemma family-control LoRA, cross-family mechanism analyses.
- Proprietary APIs (no local GPU): GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku, Claude-4.5-Sonnet, Gemini-2.5-Flash, Gemini-2.5-Pro.

Software:

- Python 3.12.13; PyTorch 2.11.0 (CUDA 13.0, cuDNN 9.19).
- `transformers` 4.57.6, `tokenizers` 0.22.2, `safetensors` 0.7.0, `sentencepiece` 0.2.1, `huggingface-hub` 0.36.2.
- LoRA training: `peft` 0.19.1, `trl` 1.4.0 (`SFTTrainer`), `datasets` 4.8.5, `accelerate` 1.13.0, `bitsandbytes` 0.49.2.
- Batched inference: `vllm` 0.21.0.
- Mechanistic analysis: `transformer_lens` 3.2.1.
- Probing + statistics: `scikit-learn` 1.7.2, `numpy` 2.3.5, `pandas` 3.0.3.
- Plotting: `matplotlib` 3.10.9.

Pre-registered controls. We pre-registered two controls before running the main experiment. The **family control** trains the identical adapter configuration (rank, learning rate, schedule, data mix) on Gemma-3-4b-it; success here indicates the recovery is architecture-agnostic rather than specific to Qwen2.5. Results are reported in App G.1 (28/28 ARB cells fixed, 16/16 SEM OOD cells fixed). The **negative control** fine-tunes the same Qwen2.5-3B-Instruct with identical hyperparameters (rank-16 attention-only LoRA, two epochs, matched batch and learning rate) on GSM8K — chain-of-thought arithmetic word problems, ∼6.7k training examples — and then evaluates on the FVQ/CVQ grid. The adapter learned its training task: GSM8K end-to-end accuracy rose from 16% to 70% on a 250-problem test split. It did not close the FVQ–CVQ gap. Table 14 reports five held-out cells (100 trials each, Wilson 95% CIs); FVQ improved modestly while CVQ *decreased* on every cell relative to baseline (post-LoRA CVQ CIs do not overlap with base). No cell reached the pre-registered fix threshold (FVQ $\geq$ 0.65 and CVQ $\geq$ 0.65); the task-specific LoRA in §5.2 reaches it on 28/28 cells. The recovery is therefore task-specific, not a general consequence of LoRA fine-tuning.

Full held-out results. Table 15 reports the full FVQ and CVQ accuracies that underlie the compact gap-only Table 3 in §5.2.

G.1 Gemma-3-4b-it family control

The family-control LoRA fits an identical adapter to Gemma-3-4b-it (grouped-query attention, 34 layers $\times$ 8 query heads with KV grouping). The training recipe matches the Qwen run: rank 16, LoRA $\alpha = 32$, attention-only $\{Q, K, V, O\}$, dropout 0.05, lr 2×10^{-4} cosine with 100-step warm-up, effective batch 64, bf16 mixed precision, gradient checkpointing. The adapter contributes 8.9M trainable parameters (0.23% of model weights). Training data is regenerated under the same $K \in \{2, 3, 5, 10\}$, $N \in \{5, 10, 15, 20\}$ grid and the same 40/40/20 FVQ/CVQ/IVQ mix; the concrete training examples differ from the Qwen run (deterministic seeding was introduced between runs).

Cell	Base FVQ / CVQ	Arith-LoRA
K=10, N=50	0.42 / 0.50	0.48 / 0.26
K=15, N=20	0.32 / 0.58	0.49 / 0.33
K=20, N=30	0.21 / 0.51	0.32 / 0.29
K=25, N=75	0.06 / 0.60	0.11 / 0.19
K=30, N=75	0.01 / 0.56	0.07 / 0.26

Table 14: Arithmetic negative-control LoRA on Qwen2.5-3B-Instruct, held-out FVQ/CVQ cells (100 trials each). Adapter learned arithmetic (GSM8K 16% → 70%) but did not recover position-indexed retrieval.

	Base			**+ LoRA**		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
(7, 30)	50%	38%	+12%	100%	100%	+0%
(7, 50)	52%	46%	+6%	100%	96%	+4%
(10, 30)	41%	46%	−5%	100%	100%	+0%
(10, 50)	42%	50%	−8%	100%	100%	+0%
(15, 20)	32%	58%	−26%	100%	98%	+2%
(15, 30)	35%	60%	−25%	100%	98%	+2%
(15, 50)	18%	59%	−41%	100%	100%	+0%
(20, 30)	21%	51%	−30%	100%	98%	+2%
(20, 50)	8%	62%	−54%	100%	96%	+4%
(20, 75)	6%	56%	−50%	100%	96%	+4%
(25, 30)	15%	59%	−44%	100%	98%	+2%
(25, 50)	5%	43%	−38%	98%	96%	+2%
(25, 75)	6%	60%	−54%	100%	93%	+7%
(30, 30)	12%	60%	−48%	100%	98%	+2%
(30, 50)	4%	54%	−50%	100%	94%	+6%
(30, 75)	1%	56%	−55%	99%	92%	+6%

Table 15: Full held-out Arbitrary-Single results for the LoRA on Qwen2.5-3B-Instruct. Same 16 cells as Table 3; FVQ and CVQ separately with the gap, before and after the adapter.

Training runs for 400 steps (1.42 epochs), reaching eval accuracy 98.3% with eval loss still improving. The model attaches to text-only inputs via the `Gemma3ForCausalLM` class with `sdpa` attention; defaults load Gemma's multimodal processor which is incompatible with our text-only data format.

Tables 16 and 17 report the held-out cell results. Across all 28 held-out Arbitrary-Single cells, both FVQ and CVQ accuracies reach 88–100% post-adapter (mean RI 99.4%, mean CVQ 97.0%). Across all 16 Semantic-Multi out-of-distribution cells, FVQ stays at 100% and CVQ reaches 83–100% (mean 96.8%). Figure 11 shows the underlying baseline failure profile across the full grid.

H Training Dynamics Data

This appendix lists the per-checkpoint mean gap (and component accuracies) that underlie Figure 4 in §6.

Method. Each checkpoint is evaluated on Arbitrary-Single with the same query format and matching rule used in the main paper (Appendix B), at $K \in \{2, 3, 5, 7\}$, $N \in \{2, 3, 5, 7, 10, 15, 20, 30\}$, 100 trials per cell. The completion-format prompt is used at every checkpoint to ensure comparability across pretraining and post-training stages; the chat-template prompt returns near-zero accuracy on early instruction-tuned SmolLM3 checkpoints (`it-mid-training` under `chat_1024` gives mean FVQ = mean CVQ = 0.001), so we report it only as a robustness comparison (Table 21). For figures that partition by load, the 32 cells split into three exhaustive bins: low ($K \leq 3$, any N; 16 cells), medium ($K \geq 5, N < 20$; 12 cells), and high ($K \geq 5, N \geq 20$; 4 cells).

SmolLM2-1.7B trajectory. Figure 12 reports the SmolLM2-1.7B trajectory under the same three-bin partition used for SmolLM3 in §6. The pattern matches SmolLM3 with a cleaner separation: low-load cells stay positive across the published four-stage curriculum (Ben Allal et al., 2025), medium-load cells hover near zero, and high-load cells stay in the reversal direction throughout pretraining and into the released final model. Table 19 reports the per-checkpoint mean FVQ, CVQ, and gap at every released checkpoint. Aggregated across all 32 cells the gap is positive at every checkpoint, with a non-monotonic curriculum trajectory — rising through early pretraining, peaking around step 625k, partially collapsing, then stabilising near the final value.

SmolLM3-3B trajectory. Table 20 reports the same statistics for SmolLM3-3B across 32 pretraining checkpoints (the three documented stages) and the four post-training stages plus the released final model. The gap is positive at every checkpoint.

Prompt-format robustness check. For the four SmolLM3 post-training stages plus the released model we additionally ran the chat-template prompt (Table 21). Magnitudes differ between prompt formats but the sign and ordering of the gap are preserved at every stage where the chat template itself does not break the model (at `it-mid-training` the chat template gives near-zero accuracy on both queries, so the gap is undefined and we omit it from Figure 4).

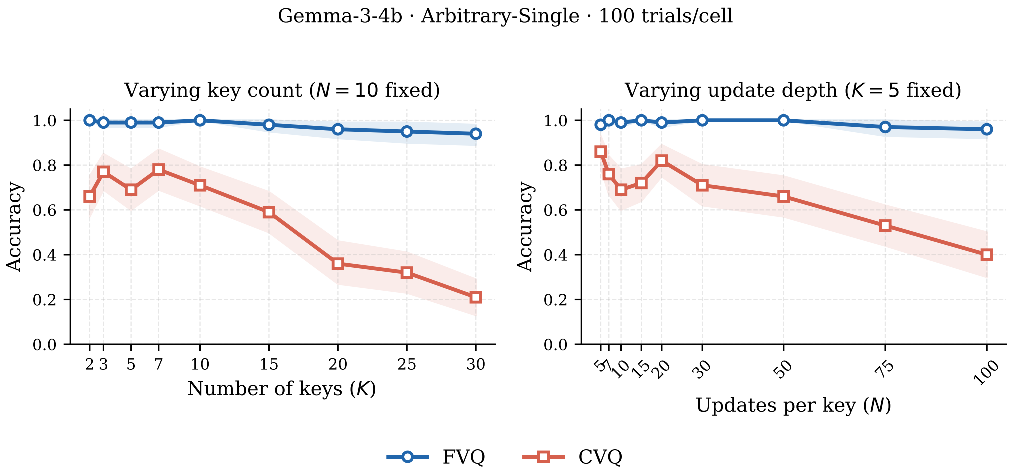

Figure 11: Baseline FVQ and CVQ accuracy on Gemma-3-4b-it across the unified grid. Clean primacy regime (FVQ $\gg$ CVQ) at every cell, in contrast to Qwen2.5-3B-Instruct which crosses into the reversal regime at high stream load. The LoRA recovery in Tables 16 and 17 closes this gap on all 28 held-out cells.

	Base			+ LoRA		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
$(7, 30)$	95%	52%	+43%	100%	100%	+0%
$(7, 50)$	95%	50%	+45%	98%	100%	−2%
$(7, 75)$	93%	31%	+62%	100%	100%	+0%
$(10, 30)$	99%	46%	+53%	100%	100%	+0%
$(10, 50)$	93%	32%	+61%	100%	98%	+2%
$(10, 75)$	92%	14%	+78%	100%	98%	+2%
$(15, 15)$	95%	44%	+51%	100%	100%	+0%
$(15, 20)$	92%	32%	+60%	100%	100%	+0%
$(15, 30)$	92%	22%	+70%	100%	100%	+0%
$(15, 50)$	85%	12%	+73%	100%	98%	+2%
$(15, 75)$	82%	9%	+73%	100%	100%	+0%
$(20, 15)$	95%	32%	+63%	98%	100%	−2%
$(20, 20)$	89%	22%	+67%	100%	100%	+0%
$(20, 30)$	83%	16%	+67%	100%	98%	+2%
$(20, 50)$	85%	8%	+77%	100%	92%	+8%
$(20, 75)$	85%	7%	+78%	100%	90%	+10%
$(25, 10)$	95%	32%	+63%	100%	100%	+0%
$(25, 15)$	88%	13%	+75%	100%	100%	+0%
$(25, 20)$	80%	12%	+68%	100%	93%	+7%
$(25, 30)$	86%	4%	+82%	98%	100%	−2%
$(25, 50)$	81%	9%	+72%	99%	91%	+8%
$(25, 75)$	—	—	—	99%	89%	+10%
$(30, 10)$	94%	21%	+73%	100%	100%	+0%
$(30, 15)$	83%	16%	+67%	100%	100%	+0%
$(30, 20)$	89%	9%	+80%	100%	98%	+2%
$(30, 30)$	91%	4%	+87%	100%	90%	+10%
$(30, 50)$	87%	6%	+81%	97%	93%	+4%
$(30, 75)$	—	—	—	96%	88%	+8%

Table 16: Gemma-3-4b-it LoRA, full held-out Arbitrary-Single results. All 28 held-out cells; FVQ and CVQ accuracy with the gap before and after the adapter. − indicates a baseline cell where the trial count was insufficient; post-adapter result is unaffected.

	Base			+ LoRA		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
$(7, 10)$	95%	83%	+12%	100%	100%	+0%
$(7, 20)$	98%	81%	+17%	100%	96%	+4%
$(7, 30)$	95%	68%	+27%	100%	100%	+0%
$(7, 50)$	87%	56%	+31%	100%	98%	+2%
$(10, 10)$	98%	80%	+18%	100%	100%	+0%
$(10, 20)$	92%	71%	+21%	100%	100%	+0%
$(10, 30)$	94%	60%	+34%	100%	100%	+0%
$(10, 50)$	86%	45%	+41%	100%	92%	+7%
$(15, 10)$	98%	66%	+32%	100%	100%	+0%
$(15, 20)$	95%	53%	+42%	100%	100%	+0%
$(15, 30)$	84%	37%	+47%	100%	96%	+4%
$(15, 50)$	72%	25%	+47%	100%	96%	+4%
$(20, 10)$	95%	48%	+47%	100%	100%	+0%
$(20, 20)$	87%	41%	+46%	100%	96%	+4%
$(20, 30)$	94%	21%	+73%	100%	91%	+9%
$(20, 50)$	78%	23%	+55%	100%	83%	+17%

Table 17: Gemma-3-4b-it LoRA, Semantic-Multi out-of-distribution results. Adapter trained only on Arbitrary-Single; 16 SEM cells share (K, N) shape with training but use multi-token semantic values. LoRA holds across this distribution shift.

Property	**Qwen2.5-3B-Inst**	**Gemma-3-4b-it**
Layers $\times$ heads	36L $\times$ 16H, dense	34L $\times$ 8H, GQA
Baseline regime	reversal (≤ -55pp)	primacy ($\leq +87$pp)
ARB held-out fixed	16 / 16	28 / 28
SEM OOD fixed	4 / 4	16 / 16
Trainable params	7.4M (0.24%)	8.9M (0.23%)
Training epochs	2.0	1.42 (early-stop)

Table 18: Cross-family LoRA recipe summary. Same recipe, opposite baseline regimes, near-identical post-LoRA accuracy.

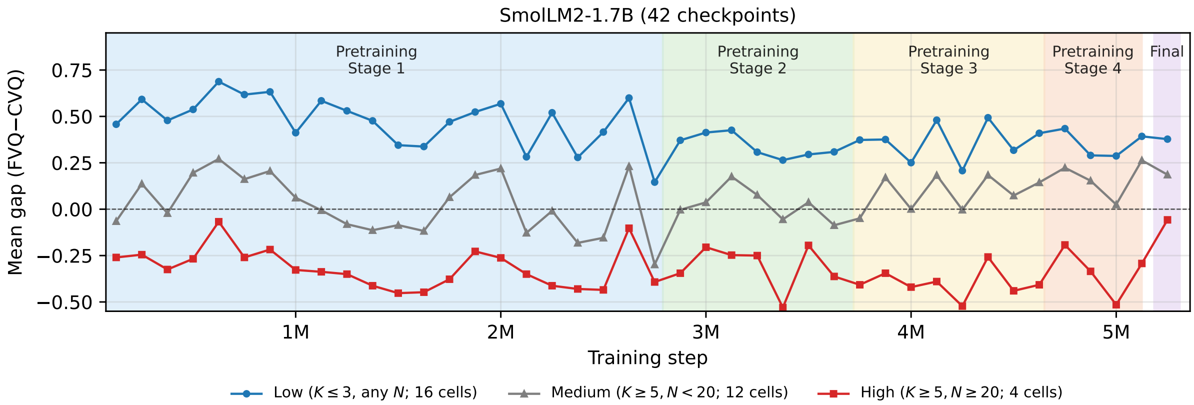

Figure 12: Mean FVQ−CVQ gap on Arbitrary-Single across the SmolLM2-1.7B (Ben Allal et al., 2025) training trajectory, under the same three-bin partition as Figure 4. Coloured bands mark the four documented pretraining stages plus the released final model.

I Cross-Family Mechanism Details (Gemma-3-4b-it)

This appendix collects the post-LoRA mechanism results on Gemma-3-4b-it underlying the cross-family replication claim in §7. Both analyses use the same merged-adapter model evaluated in App G.1; the operating point is $K{=}2, N{=}5$ for probing and three cells ($K{=}2$, $N \in \{5, 10, 50\}$) for logit lens, mirroring the Qwen analysis.

Probing classifier. Linear classifiers (sklearn logistic regression, 5-fold CV, balanced class weights) are fitted on the residual-stream activation at each of Gemma's 34 layers, separately for three labels: *condition discrimination* (RI vs CVQ query), *RI correctness* (will the model answer FVQ correctly?), and *CVQ correctness* (will the model answer the current-value query correctly?). Table 22 reports baseline-to-LoRA Δ at the eight late layers L26–L33 with parametric 95% CIs (mean $\pm 1.96 \cdot \sigma / \sqrt{5}$, pooled variance for Δ).

Logit lens. Per-trial residual-stream activations at the final layer (L33) are projected through Gemma's unembedding matrix to recover layer-level token distributions; we report $\Pr(v_{\text{first}})$ for RI and $\Pr(v_{\text{last}})$ for CVQ, averaged over 100 trials per cell with nonparametric bootstrap 95% CIs (n_boot = 2000, seed 17).

Attention routing. We replicate the §7.2 attention-routing analysis on Gemma using an HF-direct extractor (TransformerLens exceeded the available GPU memory budget on Gemma-3-4b). Operating point $K{=}2, N{=}30$, 50 trials, expA_CVQ. The top-15 promoter heads on Gemma span L14–L32 (19 of 34 layers), with Δ in $\Pr(\text{attend to } v_{\text{last}})$ ranging from $+0.33$ to $+0.47$. Per-layer mean Δ peaks at L25 ($+0.31$), with secondary peaks at L20 ($+0.26$) and L30 ($+0.24$).

Both architectures recover the FVQ–CVQ task via late-layer attention reweighting; the spatial distribution differs (Qwen concentrated, Gemma distributed), but the functional pattern — promotion of v_{last} at late layers — is shared. Paired Stage-3 ablation on Gemma is left to future work; the probing, logit-lens, and attention-routing convergences here are the cross-family evidence we report.

J Block format and LoRA recruit different internal circuits

§5.2 reports that format intervention and LoRA both close the FVQ–CVQ gap behaviourally. We test whether the two interventions share an internal locus on Qwen2.5-3B-Instruct.

Experimental design. Three conditions at $K{=}2, N{=}30$, 50 trials each (expA_CVQ): **(A)** base + Plain format, **(B)** base + Block format, **(C)** base + main LoRA + Plain format. For each of the 15 LoRA-promoter heads in L30–L33 identified in §7.2, we compute $\Pr(\text{attend to } v_{\text{last}})$ at the generation position.

Behavioural pre-check. At $K{=}2, N{=}30$ on Qwen, Block lifts CVQ from 49% to 74% (FVQ 89% $\to$ 98%; gap +40 pp $\to$ +24 pp; $n = 100$): Block produces a substantial behavioural effect at this cell, so the mechanism question is well posed.

Pooled result.

Per-head verdict. Out of 15 LoRA-promoter heads, **0 reach LoRA-level activation under**

Step	FVQ	CVQ	Gap	Cells
125,000	0.37	0.20	+0.17	32
250,000	0.50	0.18	+0.32	32
375,000	0.49	0.30	+0.19	32
500,000	0.56	0.25	+0.31	32
625,000	0.58	0.14	+0.44	32
750,000	0.58	0.24	+0.34	32
875,000	0.59	0.22	+0.37	32
1,000,000	0.55	0.37	+0.19	32
1,125,000	0.57	0.32	+0.25	32
1,250,000	0.52	0.33	+0.19	32
1,375,000	0.50	0.36	+0.14	32
1,500,000	0.53	0.45	+0.08	32
1,625,000	0.48	0.41	+0.07	32
1,750,000	0.52	0.31	+0.21	32
1,875,000	0.56	0.26	+0.30	32
2,000,000	0.57	0.23	+0.33	32
2,125,000	0.40	0.35	+0.05	32
2,250,000	0.55	0.34	+0.20	32
2,375,000	0.46	0.44	+0.02	32
2,500,000	0.52	0.43	+0.10	32
2,625,000	0.57	0.19	+0.37	32
2,750,000	0.38	0.47	−0.09	32
2,875,000	0.51	0.37	+0.14	32
3,000,000	0.53	0.33	+0.19	32
3,125,000	0.61	0.36	+0.25	32
3,250,000	0.59	0.44	+0.15	32
3,375,000	0.55	0.51	+0.05	32
3,500,000	0.58	0.44	+0.14	32
3,625,000	0.56	0.48	+0.08	32
3,750,000	0.58	0.46	+0.12	32
3,875,000	0.66	0.45	+0.21	32
4,000,000	0.58	0.51	+0.07	32
4,125,000	0.65	0.39	+0.26	32
4,250,000	0.63	0.59	+0.04	32
4,375,000	0.62	0.34	+0.28	32
4,500,000	0.60	0.47	+0.13	32
4,625,000	0.62	0.41	+0.21	32
4,750,000	0.70	0.43	+0.28	32
4,875,000	0.67	0.50	+0.16	32
5,000,000	0.64	0.55	+0.09	32
5,125,000	0.73	0.47	+0.26	32
final	0.75	0.50	+0.25	32

Table 19: SmolLM2-1.7B per-checkpoint mean FVQ, CVQ, gap on Arbitrary-Single. 32 cells per checkpoint; 100 trials/cell.

Checkpoint	FVQ	CVQ	Gap	Cells
Pretraining (completion_few_shot)				
stage1-step-40,000	0.66	0.07	+0.59	32
stage1-step-160,000	0.74	0.40	+0.34	32
stage1-step-320,000	0.86	0.33	+0.53	32
stage1-step-440,000	0.88	0.43	+0.45	32
stage1-step-600,000	0.81	0.42	+0.40	32
stage1-step-1,160,000	0.76	0.33	+0.42	32
stage1-step-1,280,000	0.72	0.45	+0.27	32
stage1-step-1,440,000	0.83	0.43	+0.40	32
stage1-step-1,560,000	0.79	0.45	+0.33	32
stage1-step-1,720,000	0.72	0.33	+0.39	32
stage1-step-1,880,000	0.62	0.35	+0.27	32
stage1-step-2,000,000	0.69	0.51	+0.18	32
stage1-step-2,160,000	0.61	0.55	+0.07	32
stage1-step-2,280,000	0.72	0.40	+0.32	32
stage1-step-2,440,000	0.66	0.60	+0.06	32
stage1-step-2,560,000	0.76	0.46	+0.30	32
stage1-step-2,720,000	0.79	0.31	+0.48	32
stage1-step-2,840,000	0.85	0.30	+0.55	32
stage1-step-3,000,000	0.79	0.51	+0.28	32
stage1-step-3,120,000	0.77	0.59	+0.17	32
stage1-step-3,280,000	0.68	0.59	+0.09	32
stage1-step-3,440,000	0.70	0.57	+0.13	32
stage2-step-3,480,000	0.75	0.43	+0.32	32
stage2-step-3,640,000	0.75	0.53	+0.22	32
stage2-step-3,840,000	0.75	0.57	+0.18	32
stage2-step-4,000,000	0.82	0.71	+0.11	32
stage2-step-4,200,000	0.80	0.55	+0.25	32
stage3-step-4,240,000	0.83	0.48	+0.35	32
stage3-step-4,360,000	0.82	0.49	+0.32	32
stage3-step-4,480,000	0.80	0.44	+0.36	32
stage3-step-4,600,000	0.82	0.49	+0.33	32
stage3-step-4,720,000	0.80	0.58	+0.22	32
Post-training (completion)				
it-mid-training	0.79	0.53	+0.25	32
it-SFT	0.82	0.63	+0.19	32
it-soup-APO	0.80	0.61	+0.19	32
it-LC-expert	0.83	0.55	+0.27	32
final	0.81	0.64	+0.16	32

Table 20: SmolLM3-3B per-checkpoint mean FVQ, CVQ, gap on Arbitrary-Single (completion format throughout). 32 cells per checkpoint; 100 trials/cell.

	Completion		Chat (1024)	
Stage	**Gap**	**Cells**	**Gap**	**Cells**
it-mid-training	+0.25	32	−0.00	32
it-SFT	+0.19	32	+0.29	32
it-soup-APO	+0.19	32	+0.20	32
it-LC-expert	+0.27	32	+0.33	32
final	+0.16	32	+0.18	32

Table 21: SmolLM3-3B post-training stages: mean gap under `completion` vs `chat_1024` prompts. The chat template returns near-zero accuracy on `it-mid-training`; the gap stays positive under both formats elsewhere.

Block (pre-registered criterion: $\Pr_B$ within bootstrap CI of $\Pr_C$ AND $\Pr_B > 0.4$). 4 of 15 are statistically indistinguishable from baseline under Block (L31H15, L32H10, L32H14, L33H14). The remaining 11 partially activate but stay well below LoRA. No head reaches LoRA's activation level.

Verdict. Block format and LoRA produce the same behavioural outcome via different internal routes. Block presumably exploits the explicit `[Round j]` headers as a discrete-step positional cue; LoRA installs attention reweighting in the L30–L33 cluster. The relationship is operational distinctness, not shared mechanism.

K MLP-only LoRA control

§5.2 fits LoRA on the four attention projections $\{Q, K, V, O\}$. A skeptical reading would observe that attention-only LoRA *can only* reweight attention, so "amplification" findings are partly struc-

Layer	Δ Cond. discr.	Δ RI correctness	Δ CVQ correctness
L26	+0.00 [+0.00, +0.00]	−0.09 [-0.18, +0.01]	**+0.11 [+0.01, +0.21]**
L27	+0.00 [+0.00, +0.00]	−0.09 [-0.18, +0.01]	**+0.15 [+0.04, +0.26]**
L28	+0.00 [+0.00, +0.00]	−0.02 [-0.11, +0.07]	**+0.15 [+0.04, +0.26]**
L29	+0.00 [+0.00, +0.00]	−0.01 [-0.09, +0.07]	**+0.18 [+0.10, +0.27]**
L30	+0.00 [+0.00, +0.00]	+0.00 [-0.08, +0.09]	**+0.18 [+0.11, +0.25]**
L31	+0.00 [+0.00, +0.00]	−0.02 [-0.10, +0.06]	**+0.17 [+0.10, +0.24]**
L32	+0.00 [+0.00, +0.00]	+0.00 [-0.07, +0.07]	**+0.14 [+0.07, +0.21]**
L33	+0.00 [-0.00, +0.01]	+0.02 [-0.08, +0.12]	**+0.15 [+0.02, +0.28]**

Table 22: Gemma-3-4b-it: Δ (LoRA − base) in probing accuracy across late layers, with 95% CIs. Bold entries exclude zero. The condition-discrimination probe saturates pre-LoRA in both architectures. RI-correctness is broadly unchanged. CVQ-correctness Δ is positive and statistically distinguishable from zero at all eight late layers, mirroring the Qwen result of Table 5.

	$P(v_{\text{first}})$ **(RI)**			$P(v_{\text{last}})$ **(CVQ)**		
(K, N)	Base	+LoRA	Δ	Base	+LoRA	Δ
$(2, 5)$	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	+0.02 [+0.00, +0.05]	0.80 [0.72, 0.87]	1.00 [1.00, 1.00]	**+0.20 [+0.12, +0.28]**
$(2, 10)$	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	+0.00 [-0.00, +0.00]	0.56 [0.47, 0.65]	1.00 [1.00, 1.00]	**+0.44 [+0.34, +0.53]**
$(2, 50)$	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	+0.00 [+0.00, +0.00]	0.59 [0.50, 0.68]	0.99 [0.99, 1.00]	**+0.41 [+0.31, +0.49]**

Table 23: Gemma-3-4b-it logit lens at L33 (final layer), baseline vs LoRA, three cells. Bold Δ entries exclude zero. The pre-LoRA $\Pr(v_{\text{last}})$ at the final layer is 0.56–0.80 across these cells (compared with 0.11 in Qwen at K=2, N=5); the post-LoRA value is 0.99–1.00 across all three cells. The starting points differ between architectures, but the post-LoRA endpoints coincide.

Property	Qwen2.5-3B-Inst	Gemma-3-4b-it
Layers × heads	36L × 16H	34L × 8H (GQA)
Top-15 promoter range	L30–L33 (4L)	L14–L32 (19L)
Max single-head Δ	+0.68 (L32H3)	+0.47 (L25H0)
Baseline suppressor cluster	L26–L30 (late)	L17–L19 + L30H6
Topology	concentrated	distributed

Table 24: Cross-architecture attention-routing comparison. Both architectures recover via late-layer attention reweighting toward v_{last}. Qwen's promoters concentrate in the last four layers; Gemma's spread across mid-to-late depth with three local peaks. The promoter topology in each architecture sits close to its baseline suppressor cluster.

Condition	Pooled Pr(attend)
A: base + Plain	0.103
B: base + Block	0.160
C: base + LoRA + Plain	0.676
Δ(B − A)	+0.056
Δ(C − A)	+0.573
Δ(B − C)	−0.516

Table 25: Pooled $\Pr(\text{attend to } v_{\text{last}})$ across the 15 LoRA-promoter heads in L30–L33. Block format barely activates the cluster (+0.056 above baseline); LoRA strongly activates it (+0.573). Block falls 51 pp short of LoRA's activation despite producing the same behavioural recovery.

tural. To probe this, we train a control LoRA targeting only MLP modules ($\{\text{gate, up, down}\}_{\text{proj}}$) on the same data.

Training. Identical configuration to §5.2: rank 16, α=32, dropout 0.05, lr 2×10^{-4} cosine, effective batch 64, bf16, on Qwen2.5-3B-Instruct. Capped at 100 optimisation steps (the attention-only LoRA took ∼ 564 steps to reach comparable eval accuracy). Trainable parameters: 22.6M (0.73% of model weights), ∼1.5× the attention-only count because $d_{\text{intermediate}} > d_{\text{model}}$ in Qwen2.5-3B. Eval accuracy reaches 97.1% at step 100.

5-cell preview. We evaluated on the five hardest cells in the held-out grid ($n = 100$ trials/cell). All five reach FVQ ≥ 96% and CVQ ≥ 94% — indistinguishable from attention-only LoRA within trial-level noise. Full 28-cell evaluation and Semantic-Multi OOD test are open.

(K, N)	**Base FVQ / CVQ**	**+MLP-LoRA FVQ / CVQ**
$(10, 50)$	42% / 50%	**98% / 96%**
$(15, 20)$	32% / 58%	**100% / 98%**
$(20, 30)$	21% / 51%	**96% / 96%**
$(25, 75)$	6% / 60%	**100% / 94%**
$(30, 75)$	1% / 56%	**96% / 96%**

Table 26: MLP-only LoRA on Qwen2.5-3B-Instruct, 5-cell preview at the hardest part of the held-out grid (100 trials/cell). All five cells close to ceiling; comparable to attention-only LoRA on the same cells (cf. Table 3).